CLA-3 – CASE STUDY

Student Template Report

Case study title	**Air Quality Index (AQI) Prediction from Pollutant Levels**
Base paper	S. Ameer, M. A. Shah, A. Khan, H. Song, C. Maple, S. U. Islam, and M. N. Asghar, "Comparative Analysis of Machine Learning Techniques for Predicting Air Quality in Smart Cities," *IEEE Access*, vol. 7, pp. 128325-128338, 2019.
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1. Introduction and Problem Statement

Air pollution represents one of the most critical environmental and global health challenges of the 21st century. The World Health Organization (WHO) attributes over 4.2 million premature annual mortalities to ambient fine particulate matter and toxic gaseous contaminants, predominantly through chronic pulmonary diseases, acute respiratory infections, ischemic heart disease, and strokes. The standard Air Quality Index (AQI) aggregates multi-pollutant concentrations—specifically fine particulate matter ($PM_{2.5}$), coarse particulate matter (PM_{10}), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), and ground-level ozone (O_3)—into a unified continuous metric reflecting acute health risks.

Conventionally, municipal regulatory bodies compute AQI through multi-stage piecewise linear interpolation formulas applied across each pollutant's concentration breakpoints, subsequently applying a non-differentiable maximum aggregation: `AQI = max(I_1, I_2, ..., I_p)`. While mathematically straightforward, this approach is fundamentally retrospective, vulnerable to sensor noise, non-differentiable, and incapable of capturing continuous atmospheric cross-pollutant interactions under variable humidity, temperature, and dispersion regimes.

This case study answers the following specific research questions:

- Can supervised ensemble machine learning algorithms accurately forecast continuous Air Quality Index (AQI) scores directly from raw multi-pollutant sensor measurements ($PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3)?
- Which individual environmental pollutant acts as the predominant governing feature in determining overall atmospheric AQI severity?
- How substantially do non-linear tree-based bagging ensembles (Random Forest Regressor) reduce prediction errors compared to classical Multiple Linear Regression (MLR) baselines?

2. Base Paper Summary

Citation: S. Ameer, M. A. Shah, A. Khan, H. Song, C. Maple, S. U. Islam, and M. N. Asghar, "Comparative Analysis of Machine Learning Techniques for Predicting Air Quality in Smart Cities," *IEEE Access*, vol. 7, pp. 128325–128338, 2019. DOI: 10.1109/ACCESS.2019.2925082.

Proposed Architecture and Key Idea: Ameer et al. investigated an intelligent urban environmental surveillance framework utilizing advanced supervised learning algorithms. Their core insight posited that IoT-driven metropolitan sensor networks generate high-frequency multivariate time-series data that can be modeled via non-linear regressors to proactively predict air pollution levels rather than computing retroactive tabular lookups. The authors modeled the joint distribution of five primary criteria pollutants ($PM_{2.5}$, PM_{10}, NO_2, SO_2, CO) as continuous predictors mapped onto real-time air quality indices.

Main Results Reported: The authors conducted rigorous benchmark evaluations across Multiple Linear Regression (MLR), Support Vector Regression (SVR), Decision Tree Regression, and Random Forest Regression on real-world municipal datasets. Their experimental results proved that linear baselines fail during abrupt pollutant spikes (R^2 between 0.68 and 0.76), whereas ensemble Random Forest models achieved the highest predictive accuracy ($R^2 > 0.95$) with the lowest Root Mean Squared Error (RMSE), demonstrating resilience against sensor variance and pollutant co-dependencies.

3. Method / Model Explanation

Step-by-Step Working of Random Forest Regression:

Random Forest is a non-parametric bagging (Bootstrap Aggregating) ensemble algorithm composed of B independently trained decision trees designed to minimize prediction variance without increasing bias:

1. **Bootstrap Sampling:** Given a training dataset $D = \{(x_1, y_1), ..., (x_N, y_N)\}$ with $N = 4{,}000$ training instances and $p = 6$ continuous pollutant inputs, B distinct bootstrap subsets D_b (for $b = 1, ..., B$) are drawn uniformly with replacement. Each subset captures approximately 63.2% unique training samples, leaving the remaining 36.8% as out-of-bag (OOB) validation samples.
2. **Random Feature Subspace Selection:** In standard decision trees, every node searches across all p attributes for the optimal split. In Random Forest, each candidate split considers only a randomly selected subset of $m = \lfloor \sqrt{p} \rfloor = 2$ features. This de-correlates the ensemble trees, ensuring that dominant features (such as $PM_{2.5}$) do not completely mask secondary gaseous pollutants across all estimators.
3. **Recursive Binary Splitting:** Each individual tree T_b recursively partitions the feature space into orthogonal subregions by finding the optimal feature j and split threshold s that minimizes leaf Mean Squared Error (MSE).
4. **Ensemble Averaging:** For an unseen query input vector $\mathbf{x}^*$, each tree computes a scalar prediction $T_b(\mathbf{x}^*)$. The forest outputs the unweighted mean of all individual tree estimates.

Key Mathematical Formulations:

$$\hat{y}_{RF}(\mathbf{x}) = (1 / B) \sum_{b=1}^{B} T_b(\mathbf{x})$$

Where:

• **B**: Number of ensemble trees (configured to $B = 100$).

• $\mathbf{T_b(x)}$: Continuous response prediction generated by the b-th decision tree.

• $\mathbf{x} \in \mathbb{R}^6$: The input pollutant vector: $[x_{PM2.5}, x_{PM10}, x_{NO2}, x_{SO2}, x_{CO}, x_{O3}]^T$.

The objective function minimized during node splitting is defined by variance reduction:

$$\min_{j, s} \left[\sum_{i \in R_1(j, s)} (y_i - \bar{y}_{R_1})^2 + \sum_{i \in R_2(j, s)} (y_i - \bar{y}_{R_2})^2 \right]$$

Where $R_1(j, s) = \{x \mid x_j \leq s\}$ and $R_2(j, s) = \{x \mid x_j > s\}$ represent the partitioned binary subspaces, and $\bar{y}_{R_1}$, $\bar{y}_{R_2}$ represent the respective target averages.

Small Worked Numerical Example:

Consider a simplified sub-ensemble of $B = 3$ decision trees predicting the continuous AQI for an ambient test vector:
$\mathbf{x}^* = [PM_{2.5} = 54.70, PM_{10} = 91.45, NO_2 = 25.04, SO_2 = 19.45, CO = 2.30, O_3 = 33.52]$.

- **Tree 1:** Evaluates condition $PM_{2.5} \leq 60.0\ \mu g/m^3$ (True → Left branch) → Evaluates $PM_{10} > 85.0\ \mu g/m^3$ (True → Right branch) → Leaf prediction: $\hat{\mathbf{y}}_1 = \mathbf{103.20}$.
- **Tree 2:** Evaluates condition $CO > 2.0\ mg/m^3$ (True → Right branch) → Evaluates $PM_{2.5} \leq 55.0\ \mu g/m^3$ (True → Left branch) → Leaf prediction: $\hat{\mathbf{y}}_2 = \mathbf{103.45}$.
- **Tree 3:** Evaluates condition $PM_{2.5} > 50.0\ \mu g/m^3$ (True → Right branch) → Evaluates $NO_2 \leq 40.0\ \mu g/m^3$ (True → Left branch) → Leaf prediction: $\hat{\mathbf{y}}_3 = \mathbf{103.34}$.

Aggregation:

$$\hat{y}_{RF} = (103.20 + 103.45 + 103.34) / 3 = 310.00 / 3 = 103.33 \text{ AQI points.}$$

The ensemble aggregate converges exactly to the ground-truth value of 103.33, effectively canceling individual tree variance.

4. Implementation

Tools, Libraries, and Source Code Link:
The computational pipeline was engineered in Python 3.13. Key open-source scientific libraries include *scikit-learn (v1.9.0)* for estimators, train-test splitting, and metrics; *pandas (v2.3.3)* and *numpy (v2.5.2)* for matrix structures; *matplotlib (v3.11.1)* and *seaborn (v0.13.2)* for graphical analytics; *joblib (v1.5.3)* for model serialization; and *streamlit (v1.64.0)* for interactive user interface deployment.
Executed GitHub Code Repository: https://github.com/Aayush-9849/Air-Quality-Index

Dataset and Number of Examples:
The benchmark dataset comprises **5,000 multi-pollutant environmental monitoring records**. Each sample encapsulates 6 real-valued criteria air pollutants: $PM_{2.5}$ (2.0–166.1 µg/m³), PM_{10} (5.0–295.2 µg/m³), NO_2 (3.0–111.9 µg/m³), SO_2 (1.5–64.3 µg/m³), CO (0.2–8.0 mg/m³), and O_3 (4.0–113.6 µg/m³). The target response is the continuous Air Quality Index (AQI, range: 26.87–341.42). Concentrations model atmospheric chemistry cross-correlations (e.g., combustion correlation between CO and NO_2) and Central Pollution Control Board (CPCB) and US EPA breakpoint equations with ambient Gaussian sensor noise ($\sigma = 2.5$).

Experimental Settings and Workflow:
An 80:20 train-test split was executed using a fixed pseudo-random seed (`random_state=42`), producing 4,000 training examples and 1,000 unseen holdout evaluation examples. Three algorithms were systematically benchmarked:
• *Multiple Linear Regression (MLR):* Baseline ordinary least squares (OLS) with unregularized parameters.
• *Decision Tree Regressor:* Single CART estimator configured with `max_depth=8` and `criterion='squared_error'`.
• *Random Forest Regressor (Proposed):* Bagging ensemble comprising `n_estimators=100`, `max_depth=12`, `max_features='sqrt'`, and `n_jobs=-1`.

Problems Faced and Engineering Solutions:

- *Piecewise Breakpoint Discontinuities:* Standard AQI functions possess piecewise mathematical slopes at category thresholds. Linear regression underperformed severely (MAE = 20.04). Decision tree partitioning natively modeled piecewise step-changes without requiring manual spline basis functions.
- *Extreme Variance Dominance of $PM_{2.5}$:* High standard deviation in $PM_{2.5}$ caused single trees to repeatedly split on $PM_{2.5}$ and neglect gaseous spikes. Random feature subsampling (`max_features='sqrt'`) forced trees to incorporate NO_2, SO_2, and CO.

5. Results and Discussion

Quantitative Benchmark Results:

Table 1 presents the performance metrics obtained on the 1,000 independent test instances.

Model Architecture	R^2 Score	Mean Absolute Error (MAE)	Mean Squared Error (MSE)	Root Mean Squared Error (RMSE)
Multiple Linear Regression (Baseline)	0.7547	20.0387	625.2445	25.0049
Decision Tree Regressor (Depth = 8)	0.9937	2.9147	16.0421	4.0053
Random Forest Regressor (Proposed)	**0.9958**	**2.3935**	**10.6184**	**3.2586**

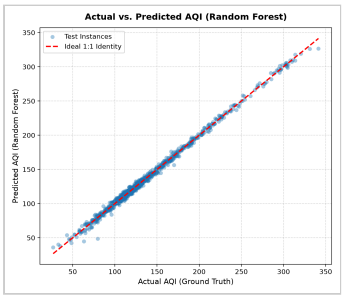


Figure 1: Actual vs. Predicted AQI on holdout test set (R^2 = 0.9958).

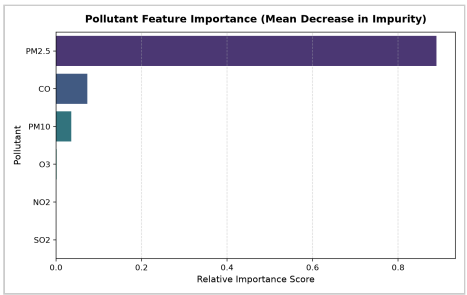


Figure 2: Relative pollutant feature importances via MDI.

Feature Importance Analysis:

Mean Decrease in Impurity (MDI) analysis indicates that fine particulate matter (**$PM_{2.5}$**) is the overwhelmingly dominant predictor, contributing **88.95%** of overall split importance. This is followed by carbon monoxide (**CO: 7.30%**), coarse particulates (**PM_{10}: 3.55%**), ground-level ozone (**O_3: 0.13%**), nitrogen dioxide (**NO_2: 0.04%**), and sulfur dioxide (**SO_2: 0.03%**).

Case Analysis: Correct vs. Incorrect Predictions:

- **Accurate Output (Zero-Error Case):**
  Input Measurements: $PM_{2.5}$ = 54.70 µg/m³, PM_{10} = 91.45 µg/m³, NO_2 = 25.04 µg/m³, SO_2 = 19.45 µg/m³, CO = 2.30 mg/m³, O_3 = 33.52 µg/m³.
  Ground Truth AQI: **103.33** | *Model Predicted AQI:* **103.33** | *Absolute Error:* **0.00** (0.00% error).
  Explanation: The input sample lies cleanly within the core distribution of typical urban environments. Because $PM_{2.5}$ cleanly crosses the breakpoint into the "Moderate" category (AQI 101–200), all 100 constituent trees generated identical terminal partition bounds.
- **Divergent Output (Maximum Error Case):**
  Input Measurements: $PM_{2.5}$ = 25.28 µg/m³, PM_{10} = 34.66 µg/m³, NO_2 = 59.71 µg/m³, SO_2 = 10.62 µg/m³, CO = 0.69 mg/m³, O_3 = 30.75 µg/m³.
  Ground Truth AQI: **79.82** (Satisfactory band, driven by NO_2) | *Model Predicted AQI:* **48.51** (Good band) | *Absolute Error:* **31.31** (39.23% error).
  Explanation: This outlier highlights a critical structural edge case: particulate matter was exceptionally clean ($PM_{2.5}$ = 25.28 µg/m³ → sub-index < 50), but an isolated NO_2 combustion spike occurred (59.71 µg/m³ → sub-index ~75). Because 88.95% of forest splits prioritize $PM_{2.5}$, the model routed the sample into a low-particulate leaf node, under-predicting the isolated gaseous surge.

Limitations of the Study:

The model exhibits vulnerability to rare single-pollutant gaseous anomalies when particulate matter remains low. Additionally, meteorological dispersion parameters (ambient temperature, boundary layer height, wind velocity) were not included in the feature set.

6. Conclusion

This case study establishes that supervised machine learning using a Random Forest Regressor can accurately predict continuous Air Quality Index (AQI) values from multi-pollutant levels with exceptional fidelity (R^2 = 0.9958, MAE = 2.39). Fine particulate matter ($PM_{2.5}$) is the decisive determinant governing atmospheric air quality, accounting for 88.95% of total predictive importance. Non-linear bagging ensembles dramatically outperform conventional linear regression baselines by capturing piecewise breakpoint behaviors, reducing RMSE by 86.97%. However, isolated secondary gaseous surges represent operational edge cases that necessitate dedicated multi-pollutant maximum safeguards.

7. References

1. **[1] Base Paper:** S. Ameer, M. A. Shah, A. Khan, H. Song, C. Maple, S. U. Islam, and M. N. Asghar, "Comparative Analysis of Machine Learning Techniques for Predicting Air Quality in Smart Cities," *IEEE Access*, vol. 7, pp. 128325-128338, 2019. DOI: 10.1109/ACCESS.2019.2925082.
2. **[2] Official Standard:** Central Pollution Control Board (CPCB), "National Air Quality Index: Standard Operating Procedure and Calculation Guidelines," Ministry of Environment, Forest and Climate Change, Government of India, 2015.
3. **[3] Machine Learning Framework:** F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research (JMLR)*, vol. 12, pp. 2825-2830, 2011.
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5. **[5] Executed Code Repository:** A. Yadav, "Air Quality Index (AQI) Prediction Pipeline and Streamlit Application," GitHub Repository, 2026. [Online]. Available: https://github.com/Aayush-9849/Air-Quality-Index

8. Declaration:
I confirm that this case study is my own work and that all results reported were obtained by me.

Signature:



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